Ballotta Sofia
Bollengier Alexis
DI Ciano Paolo
Greco Giorgio

02/12/2025

Lab 11: Regression and Classification

P1. If you want to select a set of features manually, which features would you choose and why?

Manually, I would choose features that show a high correlation with heart rate while maintaining low redundancy. I will also select physiologically meaningful variables, known as influencing variables. Also, including both time-domain and frequency-domain descriptors can help capture complementary aspects of the signals.

P2. Do the automatically selected features match your manually selected features? Explain the reasons for any similarities and/or differences.

Both the manual and automatic methods of feature selection converge on the same set of variables that are clearly correlated with heart rate. These variables include acc_mean, acc_p2p, and ppg_std, which are highly ranked by the univariate method and are clearly visible in the pair plots. The differences arise because the automatic method identifies statistical relationships that are not always apparent to us. For instance, it attributes high relevance to ppg_var and acc_std, which may seem less intuitive at first look. Manual selection tends to retain features with physiological meaning, such as ppg_mean or ppg_median, even when their statistical scores are lower. Consequently, automatic selection emphasises quantitative dependencies in the data, whereas manual selection prioritises features based on our clinical knowledge.

P3. Do you think that the feature selection was useful in this exercise? Why?

Yes, feature selection was useful in our exercise. In the notebook, using selected features consistently improved the test performance, particularly for the SVR, where the RMSE dropped from 17.15% to 11.09%. This shows that removing less informative variables helps the model generalize better and reduces noise coming from irrelevant acceleration or PPG statistics. It also makes the contribution of each feature clearer, which is important for interpreting physiological signals.

P4. How do you interpret mean, std, and rmse errors of the models?

Ballotta Sofia
Bollengier Alexis
DI Ciano Paolo
Greco Giorgio

02/12/2025

The mean error indicates whether the model consistently overestimates or underestimates heart rate and represents bias. The standard deviation quantifies precision and reflects the stability of the model across different segments or individuals. The RMSE combines both bias and variability into a single metric, providing an overall view of performance while penalising large errors.

P5. Considering all conditions, which model will you finally choose to estimate heart rate? Why?

Among all tested models, the SVR with selected features is the best choice. It achieves by far the lowest test RMSE (11.09%), the lowest test standard deviation (10.70%), and the smallest bias (–2.91%), indicating accurate, stable, and well-generalizing predictions. Compared with linear regression and SVR using all features, it shows better performance with reduced complexity and no signs of overfitting. Therefore, the SVR with feature selection offers the most reliable heart-rate estimation.

P6. If you want to select a set of features manually, which features would you choose and why?

Manually, I would select features that capture the beat-to-beat variability of the inter-beat intervals, since increased irregularity is the main indicator of atrial fibrillation. I would choose standard deviation, peak-to-peak interval (p2p), mean, median, and minimum IBI values. Standard deviation and p2p describe the overall variability and amplitude fluctuations, which are much higher in AF. Mean and median provide stable measures of the central rhythm, and the minimum value helps capture unusually short intervals that often appear during AF. These features together offer a simple but effective representation of rhythm irregularity.

P7. Do the automatically selected features match your manually selected features? Explain the reasons for any similarities and/or differences.

Yes, the automatically selected features are almost identical to my manual selection. The LASSO-based method in the notebook selected median_2, p2p, mean, std, and min, which correspond closely to the features I would choose. This happens because the dataset is composed of straightforward statistical IBI features, and AF is primarily expressed as increased variability. Both manual reasoning and the algorithm tend to prioritize features that capture this irregularity. Differences would normally arise from correlations between features or the effect of regularization, but in this case the problem is clean and the same features emerge as the most informative.

Ballotta Sofia
Bollengier Alexis
DI Ciano Paolo
Greco Giorgio

02/12/2025

P8. Do you see any signs of overfitting and/or underfitting of the models? Why?

There are clear signs of overfitting, particularly in the decision tree models. The training performance is extremely high—train accuracy close to 0.99 and train sensitivity equal to 1.00—while the test results are lower, though still strong. This gap indicates that the decision tree fits the training data almost perfectly but generalizes less effectively to unseen data, which is typical for unpruned trees. There is no meaningful evidence of underfitting, since all models achieve high performance on the test set. SVM and Naive Bayes show much smaller train-test differences, suggesting better generalization.

P9. Considering all conditions, which model will you finally choose to detect atrial fibrillation? Why?

I would choose the Support Vector Machine (SVM) as the final model. It provides high test performance while avoiding the clear overfitting seen in the decision tree. The SVM maintains a good balance between accuracy, sensitivity, and specificity, with similar results on both training and test data. Since the AF detection task involves non-linear variability patterns, the SVM's ability to model non-linear decision boundaries makes it a suitable and reliable choice. Overall, it offers the most robust generalization among the tested models.